COURSE DEMAND FORECASTING & FACULTY ALLOCATION SYSTEM

Course references another Course

Program
PK program_id
program_name
department
level
duration_years
status

Course
PK course_id
FK program_id
course_name
credits, course_level
modality
is_elective, is_popular
status, weekly_contact_hours

Prerequisite
PK prerequisite_id
FK course_id
FK required_course_id
minimum_grade
effective_from, effective_to

Faculty
PK faculty_id
faculty_name
home_department
employment_type
max_weekly_hours
status

FacultySkill
PK faculty_skill_id
FK faculty_id
skill_name
proficiency
valid_from, valid_to

StudentPlan
PK student_plan_id
student_ref
FK program_id
FK course_id
target_term
plan_status
created_at

StudentCourseHistory
PK history_id
student_ref
FK course_id
completion_term
grade

RegistrationInterest
PK registration_interest_id
FK student_plan_id (nullable)
FK course_id
term
interest_status
created_at

Availability
PK availability_id
FK faculty_id
term
status
available_hours
effective_from, effective_to

Room
PK room_id
room_name
campus
capacity
room_type
status

Section
PK section_id
FK course_id
term, scheduled_slot
FK room_id
capacity
planned_enrollment
section_status

WorkloadAssignment
PK workload_assignment_id
FK section_id
FK faculty_id
weekly_hours
assignment_status
decision_reason

UnmetDemand
PK unmet_demand_id
FK course_id
term
unmet_students
reason

PlanningEvent
PK event_id (AUTO_INCREMENT)
entity_type, entity_id
from_state, to_state
event_type
event_timestamp
actor_type
reason

1 : N

PK = Primary Key FK = Foreign Key 1:N = One-to-Many

Polymorphic audit reference:
entity_type + entity_id